CARDIOSPIKE-FPGA: COMPLETE CODEBASE DIRECTORY

A comprehensive guide to the software, hardware, and documentation files comprising the SNN-based ECG classification architecture.

1. The Python Folder (Software & Training)

This directory contains the offline data preprocessing and Backpropagation Through Time (BPTT) training scripts executed prior to FPGA deployment.

- **`preprocess.py`**
  Purpose: Reads the raw MIT-BIH database, extracts the 128-sample windows around the R-peaks, and applies Min-Max normalization (squashing voltages to an 8-bit 0-255 range).
- **`augment_data.py`**
  Purpose: Solves class imbalance by artificially injecting Gaussian noise and time shifts into rare disease classes (like F and Q), ensuring the neural network has sufficient data to learn minority arrhythmias.
- **`train_snn_sota.py` / `train_snn.py`**
  Purpose: The core PyTorch/snnTorch training script. Builds the 128 → 16 → 16 → 5 network and utilizes Surrogate Gradients to train the binary spikes via BPTT over 16 timesteps.
- **`quantize_weights.py`**
  Purpose: The bridge between software and hardware. Compresses heavy 32-bit floating-point weights into 8-bit fixed-point (Q1.7) format (.coe and .hex files) so they fit perfectly into the FPGA's physical Block RAM (BRAM).

2. The RTL Folder (VHDL Hardware)

RTL (Register-Transfer Level) contains the synthesizable VHDL code that dictates the physical logic gates on the FPGA chip.

- **`snn_pkg.vhd`** (The Package File)
  Purpose: A global configuration file defining system-wide constants (e.g., NUM_TIMESTEPS = 16) and storing the hardcoded, quantized weight arrays exported from Python.
- **`lfsr_8bit.vhd`**
  Purpose: The 8-bit Linear-Feedback Shift Register. This acts as the "Rate Encoder," generating pseudo-random thresholds to convert 8-bit ECG continuous signals into 1-bit binary spike streams without using multipliers.
- **`lif_neuron.vhd`**
  Purpose: The physical Leaky Integrate-and-Fire neuron module. Contains the digital logic to accumulate incoming weights, apply a multiplier-free bit-shift leak ($\beta = 0.9375$ via >> 4), and trigger a Zero-Reset upon firing.

- **`argmax_5.vhd`**
  Purpose: A pure combinational logic circuit that evaluates the 5 output neuron counters at the end of the inference cycle, outputting a 3-bit classification code for the winning arrhythmia class.
- **`snn_top.vhd`**
  Purpose: The Top-Level Controller Finite State Machine (FSM). It instantiates the LFSR, Neurons, and Argmax, wiring them together and managing the sequential Time-Multiplexing process across the 16 timesteps.
- **`snn_top_tb.vhd`** (The Testbench)
  Purpose: A simulation environment. Feeds synthetic ECG vectors into the top-level VHDL module to verify correct hardware behavior and timing closure before physical synthesis.

3. The Documentation (Root Folder)

The formal academic and professional documentation backing the project architecture.

- **`Proposal_Report.pdf`**
  Purpose: The formal PBEL pitch formatting the project's objectives, problem statement, and methodology.
- **`PBEL_Logbook.pdf`**
  Purpose: The 4-week structured timeline proving incremental academic progress.
- **`Literature_Review_Detailed.pdf`**
  Purpose: Maps 20 IEEE/Springer benchmark papers directly to the architectural decisions (such as the bit-shift leak and LFSR rate coding).
- **`defense_cheat_sheet.md`**
  Purpose: An end-to-end master study guide explaining the theoretical "Why" behind the mathematical and hardware tradeoffs.